



Project
Management
Learning Hub



Skill Development
Council Canada



CECOS
UNIVERSITY

Professional Certificate in Quality Management Tools and Techniques (Course Curriculum)

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Lahore, Pakistan



Course Pre-requisite	12-14 Years of Education No Experience Required
Course Duration	02 Months (Weekend + Regular Classes) 135 Credit Hours
Instructor Name	Miss Batool Akhtar

1. Course Objective

The objective of this course is to provide basic and intermediate knowledge of Project quality management tools and techniques to students and give way of working of organization that have implemented quality management system and challenges to be faced in implementing the processes.

This course aims to give a broad understanding of various concepts and techniques used in Project quality management such as quality planning, tools for quality management, quality assurance, quality monitoring and control. The course also equips the students regarding different quality standards like ISO 9000, and Six-Sigma.

It is a dedicated two-month training module in a structured and professional manner allow candidates to demonstrate following knowledge and skills:

- An understanding to use various tools, techniques and principles spelt out in Quality management and their implementation in different organizations.
- Enabling the students to contemplate the difficulties associated with the successful implementation of quality management within project environment.

Enrolling in this course will assist you in comprehending project quality management tools and techniques basics and advance features and functionality so that you can apply them to your company's project quality operations. You'll learn how to plan, control, and assure the quality of projects in order to get the most out of your capital and resource expenditures.

By taking this course, you will have a better understanding of project quality tools and techniques and functionality allowing you to make better quality management decisions. You'll learn how to efficiently monitor quality in order to maximize customer satisfaction, as well as how to monitor and control projects in order to align strategy, execution, and results.

2. Course Delivery

- a) Classroom delivery at PMLH classroom
- b) Online delivery by using “Zoom” application with real life project examples and hand on practice on MS Project.

3. Course Curriculum

Lecture Day	Topic	Detail
1	Introduction to Quality and Project quality	• What is quality? • What is project quality? • Quality Gurus
2	Quality Management System	• Quality Standards • Standards Organizations
3	Quality Cost	• Prevention Cost • Appraisal Cost • Internal Failure Cost • External Failure Cost • Cost of change
4	Quality Planning Tools	• Benchmarking • Customer Requirement • Expert Judgment
5	Quality Planning Tools	• Brainstorming • Interviews • Meetings
6	Quality Planning Tools	• Cost Benefit analysis • Prioritization Matrix • Flow charts
7	Quality Planning Tools	• Logical Data Model • Matrix Diagram • Mind Mapping • Inspection
8	Quality Planning Tools	• Affinity Diagram • Interrelationship Diagram • Multi-voting
9	Quality Planning Tools	• Nominal Group Techniques • Focus Group • Force Field analysis
10	Quality Planning Tools	• PDCA • PDSA • SDCA
11	Quality Planning Tools	• PERT • QFD • SIPOC
12	Quality Management Tools	• Checklist • Alternative analysis • Document analysis
13	Quality Management Tools	• Process analysis • Root Cause analysis • Multicriteria Decision Making
14	Quality Management Tools	• Affinity Diagram • Cause and Effect Diagram

		• Flow Charts
15	Quality Management Tools	• Histogram • Matrix Diagram • Scatter Diagram
16	Quality Management Tools	• Design for X • Problem Solving
17	Quality Control Tools	• Check Sheets • Statistical Sampling • Questionnaire and survey
18	Quality Control Tools	• Performance Review • Testing and evaluation • Control Charts
19	Quality Control Tools	• Change Request • Retrospective
20	Six Sigma Tools	• DMAIC • Waist • Value Stream mapping
21	DFSS Methodology	• IDOV • DMADV
22	FMEA	• FMEA methodology • FMEA Steps • Benefits

4. Expected Audience

This course is appropriate for anyone who is a member of a team that works on industrial projects. If at all possible, groups will be made up of organizations that complement each other in order for all participants to get as much knowledge as possible.

Our goal is to educate you on how to plan and schedule a project in the most effective way possible. Our course will teach you the planning best practices, and you will be ready to plan, control, and assure quality of your projects easily. This Certification Course will give you practical experience that will help you advance in your career.

The typical users are:

- Project Managers
- Quality Inspector
- Trades People wanting to upskill
- Contracts and Claims Support
- Support Engineer
- Quality Professional
- Quality Associate
- Implementer
- Manager
- Fresher's or early-career professionals

5. Course Outcome

- Better quality planning
- Better quality control
- Ensure quality assurance to get customer satisfaction
- Enhance quality management skills.
- Increase efficiency.
- Professional presence
- Better organize trades and materials

6. Grading System

Grade	Percentage	Performance
A	90-100%	Excellent
B	80-89%	Good
C	70-79%	Average
D	65-69%	Satisfactory
F	0-65%	Fail